A Multi-Task Evaluation of Emotion, Facial Conditions, and Deepfake Detection Towards Reliable Facial Analysis for Sri Lankan Faces

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***Abstract*— Facial-analysis models can exhibit substantial performance variation across demographic contexts, environmental conditions, prediction tasks, and unseen data distributions. This study investigates facial-analysis reliability for Sri Lankan and South Asian facial data through an integrated framework comprising facial emotion recognition (FER), age and gender estimation, facial-condition detection, and deepfake verification. FER combines SSD MobileNet V3-based face localization with CNN classification of five emotions-Angry, Fear, Happy, Neutral, and Sad-and incorporates low-light enhancement using CLAHE and gamma correction. The finalized FER model achieved 92.32% accuracy and 92.33% macro F1-score, with Happy obtaining the highest class F1-score of 96.11%. Facial-condition detection using YOLO v11n achieved 52.00% mAP@50 and 23.64% mAP@50–95, demonstrating the greater difficulty of precise localized detection. Age and gender estimation was implemented as a real-time browser-based component using Face-API.js, while its population-level quantitative validation remains limited by the available labelled data. For deepfake verification, ResNet-50, Vision Transformer (ViT), and Swin Transformer were compared under within- and cross-dataset conditions. Swin achieved the highest within-dataset accuracy of 99.83%; however, evaluation on an independent balanced dataset of 2,000 unseen images showed that ResNet-50 generalized best with 92.55% accuracy, compared with 80.95% for Swin and 69.90% for ViT. The findings demonstrate that strong within-dataset performance does not necessarily imply cross-domain reliability and emphasize the importance of task-specific, class-level, and cross-dataset evaluation when developing facial-analysis systems for locally relevant populations.**

Keywords- facial analysis, facial emotion recognition, sri lankan demographics, face-API.js, YOLOv11, deepfake detection, vision transformer.

I. INTRODUCTION

Automated facial analysis has become increasingly relevant to human-computer interaction, healthcare, security, and intelligent digital services. Advances in deep learning have enabled systems to infer facial expressions, demographic attributes, visible facial characteristics, and image authenticity from facial imagery. Large-scale datasets such as AffectNet have substantially advanced facial expression recognition (FER) [1], while modern object-detection and transformer architectures have expanded facial analysis to increasingly complex visual tasks. Anyway the high performance on established datasets does not necessarily imply comparable reliability across different populations, acquisition conditions, or unseen data distributions.

One concern is demographic generalization. Previous studies have demonstrated performance disparities across demographic groups [2], while datasets such as FairFace were introduced to improve representation across race, gender, and age [3]. Similar concerns have been identified in FER [4], [5]. These findings are particularly relevant when facial-analysis systems are applied to populations that may be underrepresented in widely used training datasets, including Sri Lankan and broader South Asian facial populations.

Environmental conditions introduce an additional source of variability. In FER, reduced illumination can suppress expression-relevant facial information and degrade recognition performance [6], [7]. The finalized FER implementation in this study therefore considers five expressions-Angry, Fear, Happy, Neutral, and Sad-under different illumination conditions. Rather than treating face detection and emotion classification as equivalent tasks, the implemented pipeline uses SSD MobileNet V3 for facial detection and localization, followed by a CNN for emotion classification. The CNN is subsequently evaluated using class-wise and aggregate performance measures to determine how reliably the five expressions are distinguished.

Reliability also extends beyond emotion recognition. Age and gender provide demographic information, while visible facial conditions require localized detection rather than whole-image classification. Furthermore, increasingly realistic manipulated facial imagery introduces an authenticity problem. Deepfake benchmarks such as FaceForensics++ and

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Celeb-DF have demonstrated that strong performance within a particular dataset does not necessarily guarantee robustness to other manipulation distributions [13], [14]. Consequently, evaluating models against unseen data is important when assessing their practical generalizability.

This study investigates these challenges through a multi-task facial-analysis framework evaluated within a Sri Lankan/South Asian context. The framework incorporates FER, browser-based age and gender estimation, YOLOv11-based facial-condition detection, and deepfake verification using CNN and transformer-based architectures. Importantly, the deepfake experiment extends beyond within-dataset evaluation by conducting cross-dataset testing on an independent balanced set of 2,000 real and manipulated images, enabling direct assessment of model generalization.

The principal contribution of this study is therefore not a single high-performing facial classifier, but a component-wise evaluation of facial-analysis reliability across multiple conditions and tasks. In particular, the study investigates (i) five-class emotion recognition under varying illumination, (ii) demographic estimation under practical acquisition conditions, (iii) localization of visible facial conditions, and (iv) within- and cross-dataset deepfake detection. This design enables performance limitations to remain visible rather than being obscured by a single framework-wide accuracy measure.

II. Related work

Existing research relevant to this study can be grouped into five areassuch as FER, demographic generalization, low-light recognition, facial-feature detection, and deepfake detection. Table I compares representative studies and highlights the limitations relevant to the present work.

TABLE I
COMPARATIVE ANALYSIS OF RELATED WORK

Research focus	**Representative studies**	**Approach / contribution**	**Limitation elevant to this study**
Facial emotion recognition	AffectNet [1]; Face2Exp [5]	Large-scale in-the-wild FER and learning strategies addressing data bias	Performance remains dependent on dataset composition and acquisition conditions
Demographic fairness	Gender Shades [2]; FairFace [3]; Xu *et al.* [4]	Evaluated demographic disparities and improved representation across age, gender, and demographic groups	Limited evidence of performance for Sri Lankan faces
Low-light FER	Xie *et al.* [6]; Wang *et al.* [7]	Enhancement, joint learning, and feature alignment for expression recognition under low illumination	Population-specific generalization remains less explored
Age and gender estimation	Levi and Hassner [8]; Rothe *et al.* [9]	CNN-based demographic attribute and apparent-age estimation	Performance can remain dependent on training population and acquisition conditions
Localized facial analysis	YOLO-based detection [11], [12]	Efficient localization and classification of visual features	Does not inherently distinguish facial conditions from expression-related interpretation
Deepfake detection	FaceForensics++ [13]; Celeb-DF [14]	Established benchmarks for manipulated facial imagery	Within-dataset performance may not represent cross-domain generalization
Transformer-based deepfake detection	ViT [15]; Swin [16]; transformer detectors [17]–[19]	Global and hierarchical representations for manipulation detection	High benchmark performance may degrade on unseen manipulation distributions

As summarized in Table I, generalization is a recurring limitation across facial-analysis tasks. Large-scale FER datasets have improved recognition under unconstrained conditions [1], while bias-oriented approaches have attempted to reduce dataset-dependent effects [5]. Nevertheless, demographic studies demonstrate that aggregate performance can conceal differences across population groups [2]-[4]. This makes population-specific evaluation particularly important when models are transferred beyond the demographic distributions on which they were originally developed.

A related issue arises from environmental variation. Low-light FER studies [6], [7] demonstrate that illumination degradation can reduce the availability of expression-relevant information and that targeted adaptation can improve robustness. Anyway demographic representation and illumination are often investigated independently. Likewise, age and gender estimation [8], [9] and localized visual-feature detection [11], [12] provide useful facial information, but each addresses only a particular dimension of facial analysis.

The generalization problem is especially evident in deepfake detection. FaceForensics++ [13] and Celeb-DF [14] established influential benchmarks for manipulated facial imagery, while ViT [15], Swin Transformer [16], and subsequent transformer-based detectors [17]-[19] have improved the modelling of manipulation patterns. Anyway strong results obtained from a model's original test distribution do not necessarily indicate equivalent performance on independently sourced manipulated imagery. This distinction motivates evaluating both within-dataset discrimination and cross-dataset generalization, rather than relying exclusively on benchmark accuracy.

The existing literature demonstrates effective solutions for individual facial-analysis tasks but reveals a broader fragmentation and generalization gap. Demographic variation, illumination robustness, localized facial characteristics, and image authenticity are commonly investigated separately, while evidence for Sri Lankan facial data remains limited. The present study addresses these limitations through a common population-focused evaluation in which the individual components retain task-specific metrics. Importantly, the deepfake component additionally incorporates an independent cross-dataset evaluation, enabling the generalization behaviour of CNN- and transformer-based architectures to be examined beyond their original test distribution.

III. Proposed Framework and Methodology

A. Overall Framework

The proposed framework integrates four complementary componentssuch as facial emotion recognition (FER), age and gender estimation, facial-condition detection, and deepfake verification. Each component addresses a distinct aspect of facial analysis and is evaluated independently using task-appropriate measures. This modular design allows differences in classification, localization, demographic estimation, and authenticity verification to remain observable rather than being represented through a single framework-wide score.

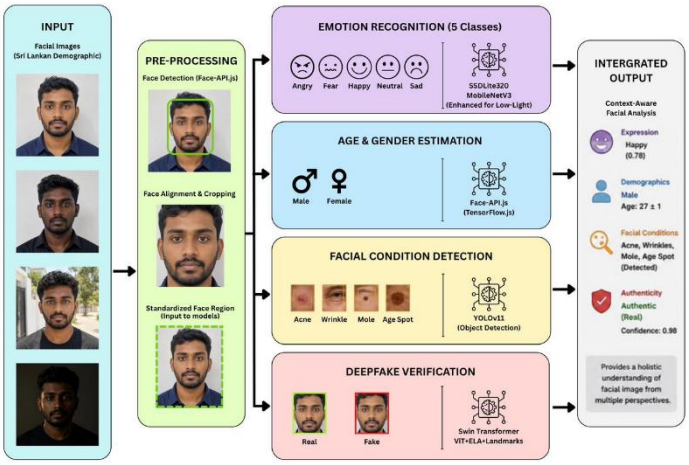


Fig 1. *Proposed Integrated Facial-Analysis Framework for Sri Lankan Demographic Faces*

B. Dataset and Experimental Setup

Task-specific datasets were used because the four components require different annotations and evaluation protocols. The FER experiments were conducted using the finalized 6,000-image dataset containing the five target expressions such as Angry, Fear, Happy, Neutral, and Sad. The data include different illumination conditions to support evaluation of emotion recognition under both normal and reduced lighting. The finalized test set used for the reported five-class CNN evaluation contained 716 images, distributed across Angry (132), Fear (140), Happy (144), Neutral (144), and Sad (156).

The age and gender component was evaluated using facial inputs captured under practical variations in illumination, camera quality, facial orientation, and background conditions. Facial-condition detection used a separately annotated image dataset in which the target conditions were represented using class-specific bounding boxes for YOLOv11 training and evaluation.

For deepfake verification, the primary experiment used the updated 6,000-image dataset, with facial images standardized to 224 × 224 pixels for model training and evaluation. Generalization was subsequently examined using an independent balanced dataset containing 2,000 previously unseen images, comprising 1,000 authentic and 1,000 manipulated samples. This separation between within-dataset and cross-dataset testing was maintained to distinguish performance on the original data distribution from generalization to independently sourced imagery.

TABLE II
DATASETS AND EXPERIMENTAL CONFIGURATION

Component	Data	Target	Experimental Conditions
FER	South Asian/local facial images	5 emotions	Normal and low light
Age/Gender	Webcam facial samples	Age, Gender	Lighting, pose, resolution, background
Facial Conditions	Annotated facial images	Facial-condition classes	Bounding-box annotations
Deepfake	Authentic + manipulated/synthetic faces	Real, Fake	Demographic and image variation

C. Facial Emotion Recognition(FER)

FER was formulated as a five-class classification task. The implemented pipeline separates face localization from emotion classification such as SSD MobileNet V3 is used to detect and localize the facial region, after which a CNN classifies the extracted face as Angry, Fear, Happy, Neutral, or Sad. This distinction is important because the two networks perform complementary functions rather than representing two interchangeable emotion classifiers. The system was evaluated under different illumination conditions, including low-light facial imagery. The CNN was trained on the updated image dataset and its performance was examined before and after the finalized adaptation/fine-tuning procedure. Evaluation considered accuracy, precision, recall, F1-score, class-wise performance, and the confusion matrix. Macro- and weighted-average measures were retained where generated by the final evaluation pipeline to account for performance across the five emotion classes. The analysis specifically examines three aspects such as the effect of illumination on recognition performance, the change following fine-tuning, and the distribution of errors among individual emotions. This permits identification of both the most reliably recognized expression and emotion pairs that remain difficult to distinguish.

D. Age and Gender Estimation

The demographic component uses Face-API.js with TensorFlow.js to perform browser-based age and gender estimation from detected facial regions. Processing is performed locally from webcam input, supporting real-time prediction without requiring continuous external-server inference. The implementation was examined under variations in lighting, camera resolution, facial orientation, and background complexity. Gender is treated as a classification output and evaluated using standard classification measures. Age is treated as a continuous estimate and assessed against available ground-truth values using prediction error rather than being represented as a categorical accuracy measure.

E. Facial-Condition Detection

Facial-condition analysis was implemented as an object-detection task using YOLOv11. Annotated facial images provide class and bounding-box information, enabling the network to identify both the type and spatial location of visible facial conditions. Evaluation uses precision, recall, F1-score, mAP@50, and mAP@50-95. Reporting both mAP measures is important because mAP@50 evaluates detection at a moderate Intersection-over-Union requirement, whereas mAP@50-95 provides a stricter assessment of localization

quality. Class-wise behaviour and confusion patterns are additionally considered when interpreting conditions that are visually similar or difficult to localize.

F. Deepfake Verification

Deepfake verification was formulated as binary classification between authentic and manipulated facial images. Three architectures-ResNet-50, Vision Transformer (ViT), and Swin Transformer, were evaluated to compare convolutional and transformer-based representations. The finalized pipeline also incorporates forensic and structural facial information developed during the project, including Error Level Analysis (ELA) and facial-landmark information. The models were first evaluated on the held-out portion of the primary dataset to establish within-dataset performance. Accuracy, precision, recall, and F1-score were recorded using the same evaluation protocol for the competing architectures. A second experiment assessed cross-dataset generalization. The trained models were evaluated without retraining on the independent 2,000-image dataset containing equal numbers of real and fake samples. This experiment was designed to determine whether strong performance on the original test distribution persists when the models encounter facial imagery from a different source. Maintaining the same trained models during this stage ensures that the resulting performance difference reflects cross-dataset transfer rather than additional adaptation to the target data.

F. Evaluation Strategy

The framework deliberately retains separate evaluation criteria for each task. Classification measures are used for FER, gender estimation, and deepfake verification; age is evaluated as a continuous prediction; and facial-condition detection additionally requires localization-based mAP measures. No Accuracy Parity Index or combined framework accuracy is calculated. This component-wise protocol is particularly important for the present study because a high classification score in one module does not establish reliability in another. The experimental design therefore enables performance to be examined across expression recognition, demographic estimation, localized facial analysis, and image authenticity, while the cross-dataset experiment provides an additional assessment of model generalization.

IV. RESULTS AND DISCUSSION

The four components were evaluated independently because they represent different prediction tasks and require different performance criteria. The results show that model reliability varies not only between tasks but also across classes, acquisition conditions, and data distributions. This is particularly evident in the FER and deepfake experiments, where aggregate performance alone does not fully describe model behaviour.

A. Facial Emotion Recognition

TABLE III CLASS-WISE PERFORMANCE OF THE FINAL FER MODEL

Emotion	Precision (%)	Recall (%)	F1-score (%)	Support
Angry	80.00	100.00	88.89	132
Fear	97.60	87.14	92.08	140
Happy	97.84	94.44	96.11	144
Neutral	95.45	87.50	91.30	144
Sad	93.55	92.95	93.25	156
Macro Avg.	92.89	92.41	92.33	716
Overall Accuracy	-	-	92.32	716

The finalized FER pipeline retained 716 successfully detected faces from the original 720-image test partition. As summarized in Table III, the CNN achieved 92.32% overall accuracy and 92.33% macro F1-score, indicating strong five-class discrimination. However, the class-level results reveal a more informative pattern than the aggregate accuracy alone. Happy was the most consistently recognized expression, whereas Angry produced the lowest F1-score. More importantly, Angry achieved perfect recall but substantially lower precision. This indicates an asymmetric classification problem: the model successfully retrieved all Angry samples, but its decision boundary for this class was comparatively permissive, causing expressions from other classes to be assigned to Angry. In contrast, Happy maintained both high precision and recall, suggesting a more distinctive learned representation. The learning curves provide additional evidence about model behaviour. Training accuracy continued to approach approximately 98%, whereas validation accuracy stabilized around 84–85%; training loss simultaneously approached zero while validation loss remained substantially higher. This divergence suggests some degree of overfitting, despite the strong final test performance. Consequently, further improvements should focus not simply on increasing training accuracy but on regularization, augmentation, and evaluation using independently collected facial images. The low-light pipeline incorporated CLAHE and gamma correction after face localization. While the final results demonstrate strong performance on the processed evaluation data, the available experiments do not constitute a controlled pre-/post-enhancement comparison using an identical test partition. Therefore, the earlier 70.91% CNN result is not interpreted as a direct improvement to 92.32%. This conservative interpretation avoids attributing the observed performance difference exclusively to illumination enhancement without a controlled ablation experiment.

B. Age and Gender Estimation

The Face-API.js component successfully demonstrated real-time browser-based age and gender inference under variations in lighting, orientation, camera quality, and background conditions. Recorded predictions showed relatively stable age estimates within individual evaluation sequences, demonstrating the functional feasibility of client-side inference. However, because the available experimental evidence does not provide a sufficiently documented labelled test population, aggregate gender classification metrics and

age-estimation error are not reported. Consequently, this component should be interpreted as a feasibility demonstration rather than evidence of demographic accuracy or fairness. A larger labelled Sri Lankan evaluation population is required before population-level performance claims can be made.

C. Facial-Condition Detection

TABLE IV FACIAL-CONDITION DETECTION PERFORMANCE

Metric	Final Performance (%)
Precision	70.83
Recall	46.91
mAP@50	52.00
mAP@50–95	23.64

Table IV demonstrates a pronounced imbalance between precision and recall. The comparatively higher precision indicates that predictions made by YOLO11n were often credible; however, the substantially lower recall shows that the primary weakness was missed facial conditions rather than excessive false detections. This distinction is practically important because improving precision alone would not address the detector's principal limitation. A second limitation emerges from the decline between mAP@50 and mAP@50–95. Performance at the stricter IoU range is less than half the mAP@50 value, indicating that accurate spatial localization is considerably more difficult than approximate detection. Small affected regions, weak visual boundaries, and similarity between facial textures may contribute to this behaviour, although dedicated error analysis would be required to determine their individual effects. The result therefore suggests that future improvement should prioritize localization quality and missed-instance reduction, rather than treating the detector's overall performance as a single optimization problem. Additional high-quality bounding-box annotations, greater representation of small conditions, and targeted augmentation are likely to be more valuable than merely extending training duration.

D. Deepfake Verification

TABLE V WITHIN- AND CROSS-DATASET DEEPFAKE DETECTION PERFORMANCE

Model	Evaluation	Acc. (%)	Prec. (%)	Recall (%)	F1 (%)
ResNet-50	Within	98.50	98.54	98.50	98.50
	Cross	92.55	93.52	92.55	92.51
ViT	Within	95.33	95.37	95.33	95.33
	Cross	69.90	70.93	69.90	69.52
Swin	Within	99.83	99.83	99.83	99.83
	Cross	80.95	86.21	80.95	80.23

Table V reveals a clear distinction between source-distribution performance and cross-dataset robustness. All three architectures performed strongly on the original test distribution, with Swin producing the highest within-dataset result. However, the ranking changed when the same trained models were exposed to 2,000 independently sourced images without retraining. ResNet-50 retained the strongest performance under this distribution shift. Its accuracy decreased by only 5.95 percentage points, compared with reductions of 18.88 points for Swin and 25.43 points for ViT. Thus, the approximately 1.3-point within-dataset advantage of Swin over ResNet-50 did not translate into superior robustness; instead, ResNet-50 exceeded Swin by 11.60 percentage points on the unseen dataset. This reversal is more informative than the near-perfect source-test scores themselves. If architecture selection had been based exclusively on within-dataset accuracy, Swin would have been selected as the strongest detector. Cross-dataset evaluation instead identifies ResNet-50 as the more reliable model under the evaluated distribution shift. The result therefore demonstrates that model ranking itself can be dataset-dependent, not merely the absolute performance value. ViT showed the largest degradation, retaining only 69.90% cross-dataset accuracy despite achieving 95.33% within the source distribution. This suggests stronger sensitivity to the change in data distribution in the present experiment. Nevertheless, the current results do not establish which learned features caused the different generalization behaviour. Feature-attribution or explainability experiments would be required before attributing ResNet-50's robustness to specific local artifacts or representations. These results directly support the study's broader argument: near-perfect benchmark performance should not be interpreted as evidence of deployment-level reliability without evaluation on independently sourced data. This is particularly important for deepfake detection, where manipulation methods, compression, image sources, and preprocessing pipelines can vary substantially across datasets [13], [14].

E. Overall Discussion

The component-level results reveal three different forms of reliability limitation. First, FER demonstrates that strong aggregate classification performance can conceal class-specific error asymmetry, as observed for Angry. Second, facial-condition detection shows that classification confidence and localization reliability are distinct concerns; comparatively acceptable precision did not prevent substantial missed detections or imprecise localization. Third, and most importantly, the deepfake experiment demonstrates that performance ranking can change under dataset shift, with the strongest source-distribution model no longer being the strongest model on unseen data. These findings collectively argue against evaluating heterogeneous facial-analysis systems using a single notion of "accuracy." Reliability depends on the task being performed and the conditions under which the model is evaluated. For FER, class-wise errors are important; for facial-condition detection, both identification and localization must be considered; and for deepfake verification, cross-dataset robustness becomes critical. The findings should also be interpreted within the scope of the available data. The study does not establish demographic fairness, nor does it demonstrate that facial conditions causally influence FER errors. Instead, it provides evidence that population context, acquisition conditions, task characteristics, and dataset distribution should remain explicit dimensions of facial-analysis evaluation. This component-wise interpretation provides a more defensible basis for assessing locally relevant facial-analysis systems than relying on a single framework-wide performance score.

V. Conclusion and Future Work

This study investigated the reliability of facial analysis for Sri Lankan and South Asian facial data through an integrated framework comprising facial emotion recognition, age and gender estimation, facial-condition detection, and deepfake verification. Rather than representing these heterogeneous tasks through a single performance measure, each component was evaluated independently using task-appropriate criteria, allowing differences associated with expression class, localization difficulty, acquisition conditions, and dataset distribution to remain visible. The FER pipeline, combining SSD MobileNet V3-based face localization with CNN-based five-class emotion classification, achieved 92.32% test accuracy and 92.33% macro F1-score. Class-level evaluation nevertheless revealed variation among expressions, with Happy producing the highest F1-score and Angry the lowest. The YOLO11n facial-condition detector achieved 52.00% mAP@50 and 23.64% mAP@50–95, indicating that precise localization of facial conditions remained considerably more challenging. The browser-based age and gender component demonstrated practical real-time inference, although the available evaluation evidence was insufficient to establish population-level demographic accuracy.

The most significant finding emerged from the deepfake experiments. Although Swin Transformer achieved the highest within-dataset accuracy of 99.83%, cross-dataset evaluation produced a different model ranking. On 2,000 previously unseen real and manipulated images, ResNet-50 achieved the strongest accuracy of 92.55%, compared with 80.95% for Swin Transformer and 69.90% for ViT. This result demonstrates that superior performance on a source test distribution does not necessarily correspond to stronger generalization and highlights the importance of independent cross-dataset evaluation when assessing deepfake detectors. The findings demonstrate that facial-analysis reliability is task-, class-, condition-, and distribution-dependent. The study does not establish the elimination of demographic bias; rather, it provides a population-focused experimental basis for examining the behaviour and limitations of complementary facial-analysis functions in a locally relevant setting.

Future work should expand the scale and demographic coverage of labelled Sri Lankan facial datasets and conduct controlled evaluation across illumination, pose, camera quality, age, gender, and other relevant acquisition factors. FER experiments should further investigate class-specific confusion under standardized low-light conditions, while facial-condition detection requires additional annotated data and improved localization of small or visually ambiguous regions. The age and gender component should be validated against a labelled test population to quantify prediction error and demographic performance. For deepfake detection, future studies should extend cross-dataset evaluation across multiple manipulation techniques and datasets and employ explainability or feature-attribution methods to investigate why particular architectures exhibit greater robustness under distribution shift.

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